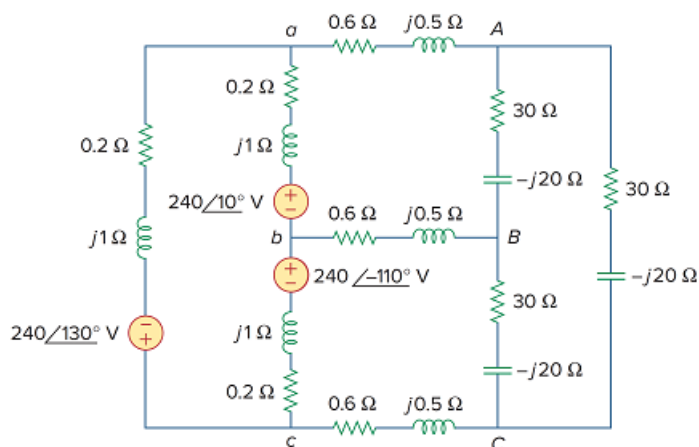


Problem

A balanced three-phase circuit is shown in Fig. 12.70 on the next page. Use *PSpice* or *MultiSim* to find the line currents I_{aA} , I_{bB} , and I_{cC} .

Figure 12.70



Step-by-step solution

Step 1 of 4

Due to delta – connected source we follow example 12.12. We type total pts = 1. Start frequency = 0.1592, & an End frequency = 0.1592. The schematic is shown below. After it is saved & simulated, we obtain An output file which includes,

Step 2 of 4

<i>Frequency</i>	<i>1M(V_PRINT1)</i>	<i>1P(V_PRINT1)</i>
1.592E-01	6.5810E+00	9.866E+01
<i>Frequency</i>	<i>1M(V_PRINT2)</i>	<i>1P(V_PRINT2)</i>
1.592E-01	1.140E+01	-1.173E+02
<i>Frequency</i>	<i>1M(V_PRINT3)</i>	<i>1P(V_PRINT3)</i>
1.592E-01	6.581E+00	3.866E+01

Step 3 of 4

Thus,

$$\begin{aligned}
 I_{aA} &= 6.581 \angle 98.60^\circ \text{ A} \\
 I_{bB} &= 11.4 \angle -113^\circ \text{ A} \\
 I_{cC} &= 6.581 \angle 38.66^\circ \text{ A} .
 \end{aligned}$$

